Some Results on Reconfiguration Graphs of Independent Sets

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Research Seminar at Kyutech Algorithms Group

Outline

Introduction

- Combinatorial Reconfiguration
- Independent Sets
- Reconfiguration Graphs of Independent Sets

Some Questions and Results

- Observations
- Typical Questions
- Trees and Cycles

... is the study of *reconfiguration graphs*.

Typical Properties

- *adjacency* is polynomial testable.
- *reconfiguration graph* is huge.

Reconfiguration v.s. Solution Space

- *configurations* $\equiv$ *feasible solutions* of a problem.
- *reconfiguration rule* $\equiv$ small change that preserves the "feasibility" of a solution.

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Examples of Reconfiguration Problems:

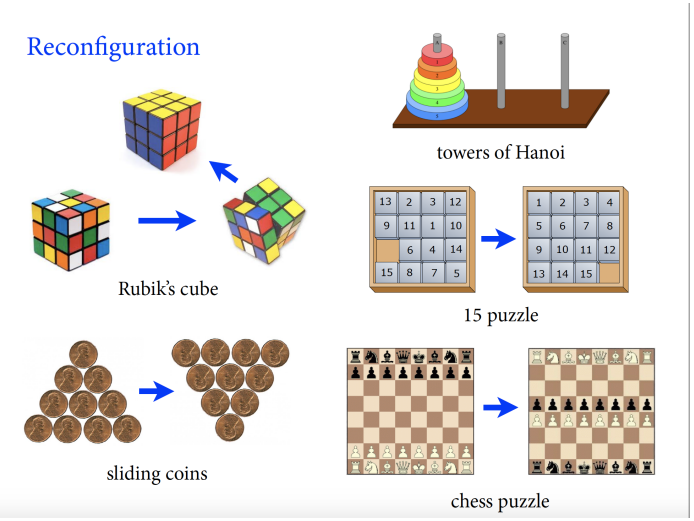

Figure: Games and Puzzles. [Anna Lubiw, CoRe2019]

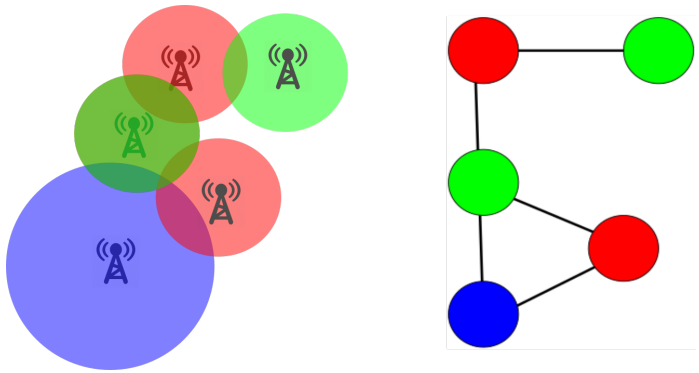

Figure: Frequency Re-Assignment and Vertex-Recoloring.

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An *independent set* of a graph $G = (V, E)$ is a vertex-subset $I \subseteq V$ such that for any pair $u, v \in I$, *no* edge connects u and v.

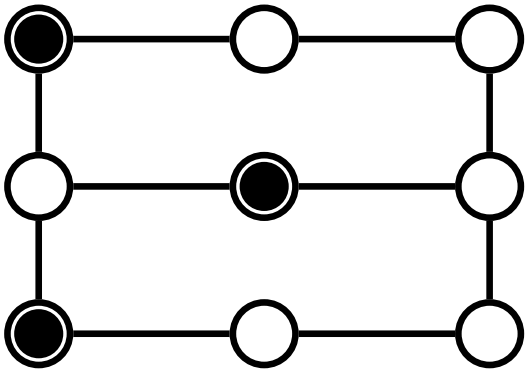

Figure: Example of an independent set whose members are depicted with black tokens.

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Let $G = (V, E)$ be a graph on n vertices.

Under Token Sliding: $\mathsf{TS}_k(G)$

- Each *node* is an independent set of size k.
- Two nodes I, J are *adjacent* if there exist $x, y \in V$ such that $I \setminus J = \{x\}$, $J \setminus I = \{y\}$, and $xy \in E(G)$.

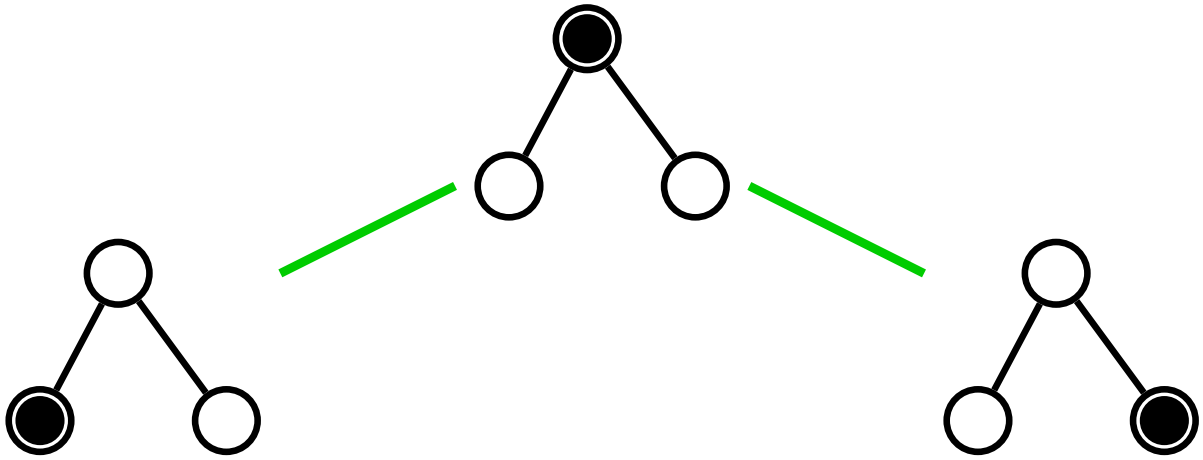

Figure: $\mathsf{TS}_1(P_3)$.

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Let $G = (V, E)$ be a graph on n vertices.

Under Token Jumping: $\mathsf{TJ}_k(G)$

- Each *node* is an independent set of size k.
- Two nodes I, J are *adjacent* if there exist $x, y \in V$ such that $I \setminus J = \{x\}$, $J \setminus I = \{y\}$, ~~and $xy \in E(G)$~~.

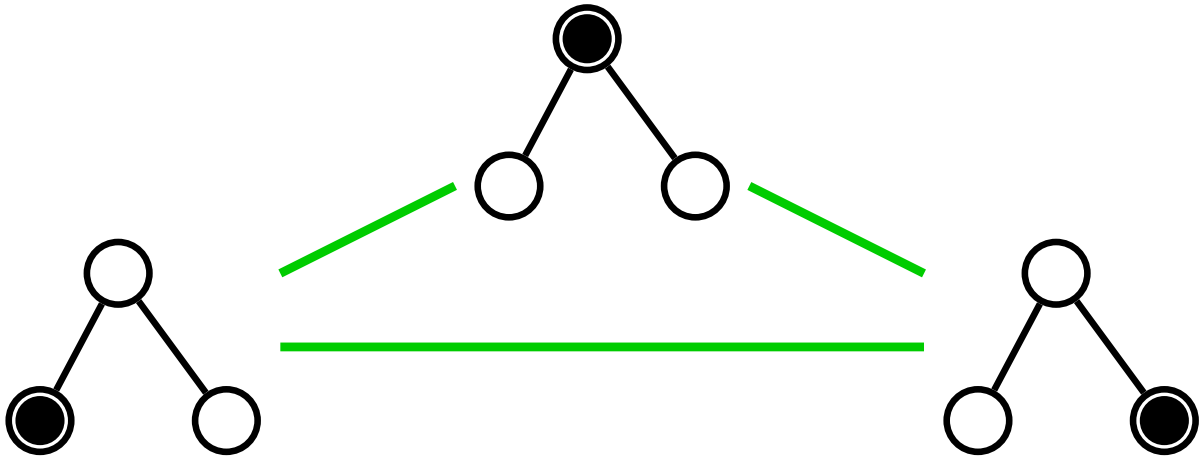

Figure: $\mathsf{TJ}_1(P_3)$.

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Let $G = (V, E)$ be a graph on n vertices.

Under Token Addition/Removal: $\mathrm{TAR}_k(G)$

- Each *node* is an independent set of size *at least* k.
- Two nodes I, J are *adjacent* if there exists $x \in V$ such that $I \Delta J = (I \setminus J) \cup (J \setminus I) = \{x\}$.

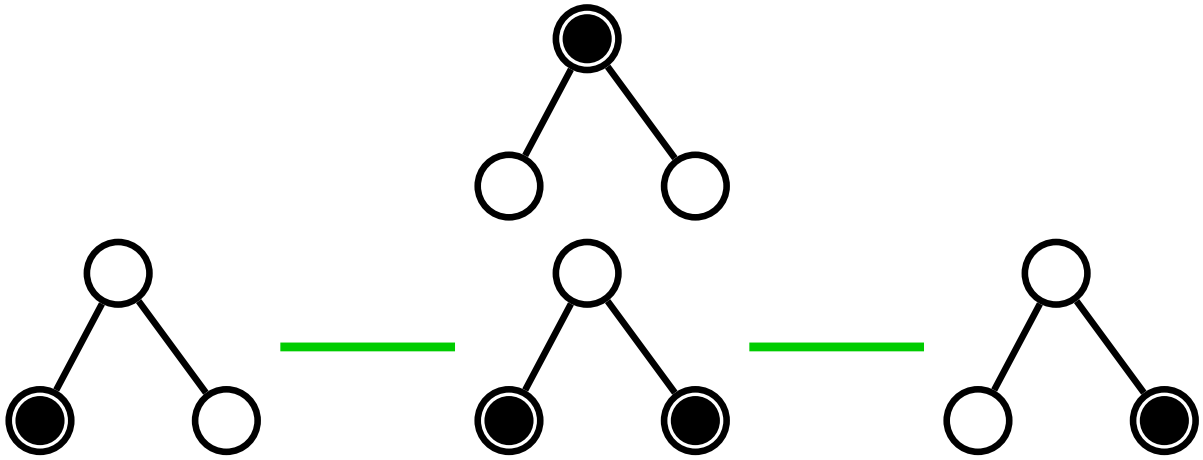

Figure: $\mathrm{TAR}_1(P_3)$.

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Let $G = (V, E)$ be a graph on n vertices. Recall that two graphs G, H are *isomorphic*, denoted by $G \simeq H$, if there is a bijection $f : V(G) \to V(H)$ such that for any pair $u, v \in V(G)$, $uv \in E(G)$ if and only if $f(u)f(v) \in E(H)$.

- For two independent sets I, J of size k, there is a path between them in $\mathsf{TJ}_k(G)$ if and only if there is a path between them in $\mathsf{TAR}_{k-1}(G)$ (see [Kamiński et al. 2012]).

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- By definition, $\mathsf{TS}_k(G)$ is a *spanning subgraph* of $\mathsf{TJ}_k(G)$.

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- By definition, $\mathsf{TS}_k(G)$ is a *spanning subgraph* of $\mathsf{TJ}_k(G)$.
- If $k = \alpha(G)$—the *maximum* size of an independent set of G—then $\mathsf{TS}_k(G) \simeq \mathsf{TJ}_k(G)$. If $k = 1$, then $\mathsf{TS}_k(G) \simeq G$ and $\mathsf{TJ}_k(G) \simeq K_n$.

Some Questions and Results

Typical Questions

Let $G = (V, E)$ be a graph on n vertices. Suppose that $\mathsf{R} \in \{\mathsf{TS}, \mathsf{TJ}, \mathsf{TAR}\}$. Typically, one may ask

1. Given two nodes I, J of $\mathsf{R}_k(G)$, is there a *(shortest) path* between them? If yes, can we find it efficiently?
2. Is $\mathsf{R}_k(G)$ *connected*?
3. Does $\mathsf{R}_k(G)$ belong to some *graph class* $\mathcal{G}$?
4. Does there exist a graph G that is *isomorphic* to $\mathsf{R}_k(G)$, or, at least, in the *same graph class* as $\mathsf{R}_k(G)$?
5. and so on.

Note: Alikhani and Fatehi [Alikhani and Fatehi 2017] initiated the study of Question 3 for $\mathsf{TAR}_k(G)$. In this talk, we will focus on *finding some structural properties of* $\mathsf{TS}_k(G)$ when G is either a tree or a cycle.

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Lemma

Let T_n and C_n be respectively a tree and a cycle on n vertices. Then,

(a) $\mathsf{TS}_k(T_n)$ *is bipartite.*

(b) $\mathsf{TS}_k(C_n)$ *is bipartite when n is even; otherwise, it is not. Moreover, when n is odd and $k = \alpha(C_n) = \lfloor n/2 \rfloor$, we have $\mathsf{TS}_k(C_n) \simeq C_n$.*

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Hint.

Let $I = \{v_1, \ldots, v_k\} \in V(\mathsf{TS}_k(G))$; and token t_i be on v_i. Suppose that there is a cycle $\mathcal{C}$ in $\mathsf{TS}_k(G)$ containing I.

Observation

To form $\mathcal{C}$, tokens must be moved based on some *permutation* π of $v_1, \ldots, v_k$, i.e., *token t_i on v_i will finally move to $\pi(v_i)$*.

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Hint (cont.)

(a) In a tree T_n, the only way to slide t_i from v_i to $\pi(v_i)$ is through the unique $v_i\pi(v_i)$-path. There are two types of token-slides:

(1) blue token-slides: *along* the $v_i\pi(v_i)$-path;
(2) red token-slides: *outside* the $v_i\pi(v_i)$-path.

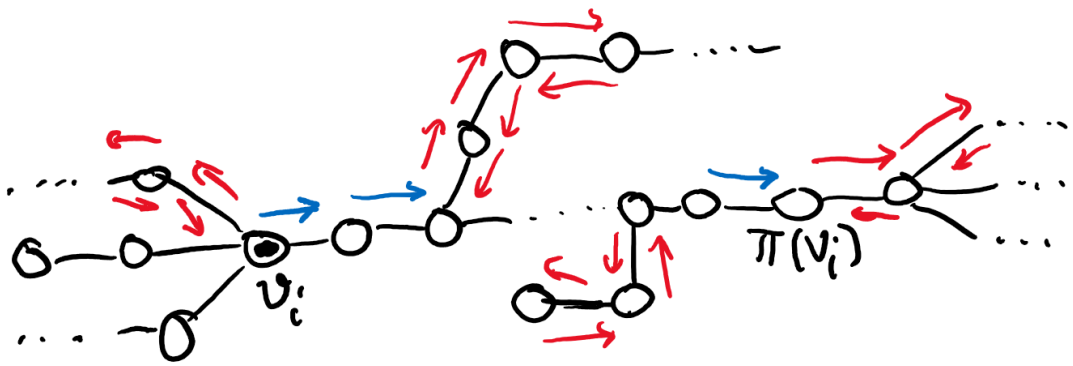

Figure: Possible types of token-slides when moving t_i.

To form $\mathcal{C}$, we need $2\sum_{i=1}^{k} \text{dist}_{T_n}(v_i, \pi(v_i))$ blue $+$ some even number of red token-slides.

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Hint (cont.)

(b) In a cycle C_n, the permutation π defined by $\pi(v_i) = v_{i+1}$ for $i \in \{1, \ldots, k-1\}$ and $\pi(v_k) = v_1$ can be used repeatedly to generate any $\mathcal{C}$ (if it exists).

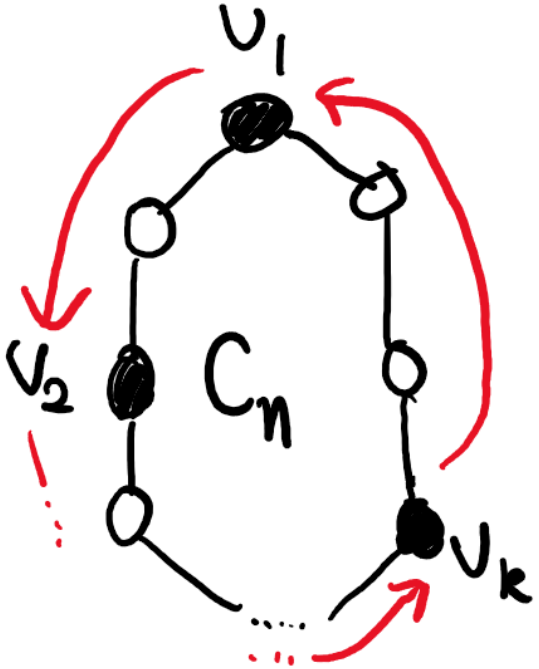

Figure: The permutation π helps generating any cycle $\mathcal{C}$ in $\mathrm{TS}_k(C_n)$ containing I. The arrows indicate next targets.

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A graph is *Eulerian* if it is connected and has a closed walk containing all edges. Let $C_n = w_1 \dots w_n w_1$.

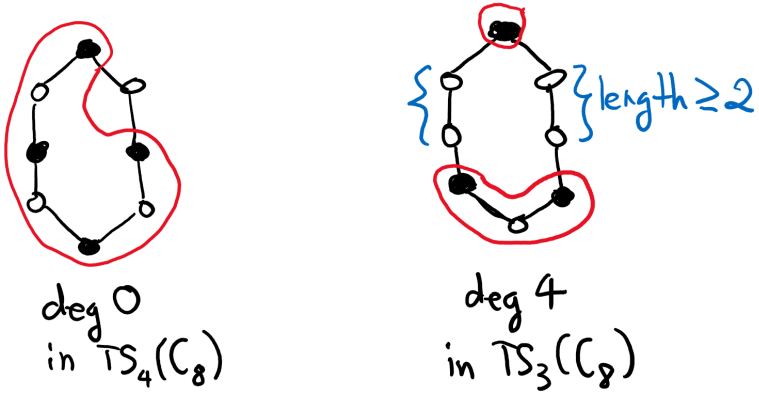
Lemma

$\mathsf{TS}_k(C_n)$ *is Eulerian.*

Hint.

Note that a graph is Eulerian if and only if every vertex has even degree. Let $I \in \mathsf{TS}_k(C_n)$. The cycle C_n contains *token-paths* and *empty-paths*.

- token-paths: $w_i w_{i+1} \dots w_j$, where $w_i, w_{i+2}, w_{i+4}, \dots, w_j$ contain tokens.
- empty-paths: no tokens + length ≥ 2.



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Question

Can we find

- a graph G, and
- a property $\mathcal{P}$,

such that *G and $\mathsf{TS}_k(G)$ both satisfy $\mathcal{P}$*?

- G is either T_n or C_n, and $\mathcal{P}$ is "being bipartite".
- G is C_n, and $\mathcal{P}$ is "being Eulerian".
- Another example?

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- G is C_n, and $\mathcal{P}$ is "being Eulerian".
- Another example?

Question

The *girth* of a graph is the size of its *smallest cycle*. If a graph has no cycle, its girth is ∞. What about girth$(\mathsf{TS}_k(T_n))$?

- From the above Lemma, girth$(\mathsf{TS}_k(C_n))$ is either n or ∞.

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Lemma

Let P_n and C_n be respectively a path and a cycle on n vertices. Then, for n large enough, and some k

(a) $\mathsf{TS}_k(P_n)$ *is not planar. Thus, so is* $\mathsf{TJ}_k(P_n)$.

(b) $\mathsf{TS}_k(C_n)$ *is not planar. Thus, so is* $\mathsf{TJ}_k(C_n)$.

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(b) $\mathsf{TS}_k(C_n)$ *is not planar. Thus, so is* $\mathsf{TJ}_k(C_n)$.

Hint (based on comments from David Avis-sensei, KyotoU).

Consider the graph G_n whose components are $\mathsf{TS}_k(P_n)$, for $k \in \{0, 1, 2, \ldots, \alpha(G)\}$.

(a) Let $f(n) = |V(G_n)|$ and $g(n) = \sum_{I \in V(G_n)} \deg_{G_n}(I)$. We have

- $f(n) = f(n-1) + f(n-2)$ for $n \geq 2$; $f(0) = 1$; $f(1) = 2$.
- $g(n) = g(n-1) + g(n-2) + 2f(n-3)$ for $n \geq 3$; $g(0) = 0$; $g(1) = 0$; $g(2) = 2$.

If G_n is planar, we must have $g(n) \leq 2(3f(n) - 6)$ for every n. This does not hold when $n \geq 19$. [Please verify!]

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Hint.

Consider the graph G_n whose components are $\mathsf{TS}_k(C_n)$, for $k \in \{0, 1, 2, \ldots, \alpha(G)\}$.

(b) Take a vertex v of C_n. For each k, the subgraph of $\mathsf{TS}_k(C_n)$ induced by nodes (independent sets) that do *not* contain v is indeed isomorphic to $\mathsf{TS}_k(P_{n-1})$. Note that every subgraph of a planar graph must be also planar.

□

Note: Consequently, there is a *maximal outerplanar graph* G such that $\mathsf{TS}_k(G)$ and $\mathsf{TJ}_k(G)$ are both *non-planar* for some k.

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Question

For which *value* of k, $\mathsf{TS}_k(P_n)$ is *planar*, for any n?

- Trivially, $k = 1$ or $k = \lfloor n/2 \rfloor$. Any non-trivial bound?

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Question

Can we find a *graph* G such that $\mathsf{TS}_k(G)$ is *planar*?

- Trivially, one can take a *star graph* $K_{1,n}$ (i.e., a tree formed by joining n leaves to a center vertex.).

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- Trivially, one can take a *star graph* $K_{1,n}$ (i.e., a tree formed by joining n leaves to a center vertex.).
- **My guess:** G can be a *double-star graph* (i.e., a tree formed by joining the centers of two disjoint stars). Moreover, when G is double-star, $\mathsf{TS}_k(G)$ is a tree.

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- **My guess:** G can be a *double-star graph* (i.e., a tree formed by joining the centers of two disjoint stars). Moreover, when G is double-star, $\mathsf{TS}_k(G)$ is a tree.
- **More general guess:** We knew that $\mathsf{TS}_k(G)$ is non-planar for some k when G is a "long" path. Thus, to make it planar, we might take G as a *tree* with some *small diameter* d?

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